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First Assessment of ICESat-2 Summer Sea Ice Freeboard With Airborne Validation

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Key Points:

- We evaluate July 2022 airborne LiDAR data and imagery over ~50 km of sea ice with near-coincident ICESat-2 data
- Sea ice melt ponds are not explicitly classified by ICESat-2 but are rarely used as sea surface height reference in freeboard retrieval
- Coregistered surface heights show strong agreement ($R \geq 0.9$) and freeboard differences range from -0.05 ± 0.12 m to 0.01 ± 0.10 m

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Abstract ICESat-2 has demonstrated solid performance in retrieving sea ice freeboard during winter, when sea ice surface types are well-defined and leads provide reliable sea surface height references. Summer retrievals are challenging because melt ponds are not explicitly distinguished in the ATL07 surface classification and may be identified as ice or leads. Here we assess ICESat-2 summer sea ice observations using airborne LiDAR and imagery collected along near-coincident ground tracks during a July 2022 calibration and validation campaign. Over a ~50-km Arctic sea ice transect, we compare ICESat-2 surface classifications, elevations, and freeboards with high-resolution airborne measurements. Melt ponds are generally not selected as sea surface height references in the ICESat-2 freeboard retrieval. ICESat-2 and airborne surface heights agree well ($R \geq 0.9$) and there are minimal freeboard differences (-0.05 ± 0.12 m to 0.01 ± 0.10 m). ICESat-2 summer freeboard retrievals are largely unaffected by melt pond misclassification, supporting the potential for year-round sea ice thickness monitoring.

Plain Language Summary Over the polar seas, NASA's Ice, Cloud, and Land Elevation Satellite (ICESat-2) is used to estimate sea ice thickness from surface height measurements. Ice thickness estimates are most reliable in autumn-spring when sea surface heights are easily identifiable. In summer, the accuracy of ice thickness estimates is unknown because surface melt complicates the sea surface height retrieval. Puddles of meltwater form on the ice surface that sometimes appear like the ocean, making it difficult to distinguish the sea surface height needed to estimate ice thickness. In July 2022, an aircraft collected measurements coincidentally with ICESat-2 with the goal of evaluating ICESat-2's summertime retrievals. Comparisons revealed that ICESat-2 sea ice heights show strong agreement with airborne measurements, with a correlation greater than 0.9. We also found that melt ponds are generally not used as an ocean reference. These results give confidence for reliable ICESat-2 measurements year-round.

1. Introduction

The Arctic sea ice cover is responding to anthropogenic climate change (Lee et al., 2023). Its thickness (Kacimi & Kwok, 2024; Kwok, 2018; Sumata et al., 2023) and extent (Parkinson, 2022) have decreased since the late 1970s, and Arctic sea ice has shifted from a multiyear to seasonal ice regime (Babb et al., 2023; Maslanik et al., 2011). As a result of climate feedbacks (Goosse et al., 2018), sea ice coverage has declined in all months (Parkinson, 2022). The loss of sea ice has important consequences for Arctic coastal communities, ecosystem services, marine traffic, and other socioeconomic aspects (Meier et al., 2014; Post et al., 2013). Thus, it is imperative to monitor how, when, and where sea-ice loss is occurring. With the advent of satellite altimetry, the retrieval of sea ice thickness has been a quantum leap forward in sea ice monitoring capabilities. Sea ice thickness is a key metric linked to sea ice survivability (e.g., Day et al., 2014; Zhou et al., 2022), which makes it a critical indicator of the Arctic sea ice state. In particular, summertime sea ice thickness retrievals can unveil crucial aspects of atmosphere-ice-ocean feedbacks in a season when sea ice loss is greatest. While major advances in summer ice thickness retrievals have been made using radar altimetry data (Landy et al., 2022), the capability of laser altimetry data for summertime retrievals is wholly uncertain. This work takes the first step in evaluating laser altimetry data of melting sea ice to determine its capability of accurate summertime retrievals.

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The launch of NASA's Ice, Cloud, and land Elevation Satellite-2 (ICESat-2) in 2018 marked a significant advancement in our ability to monitor Earth's polar regions (Neumann et al., 2019). ICESat-2 provides dense along-track sampling and high vertical precision, enabling detailed measurements of sea ice freeboard, ice sheet elevation, and vegetation structure. Over sea ice, ICESat-2's laser altimeter, the Advanced Topographic Laser Altimeter System (ATLAS), offers a unique tool for tracking changes in ice thickness and surface roughness, contributing to long-term records of Arctic and Antarctic sea ice variability (Kwok, Markus, et al., 2019).

ICESat-2's mission science requirements specify that the freeboard must be determined with an uncertainty of ≤3 cm (Markus et al., 2017), which has been shown to be met in winter and spring in multiple validation studies (e.g., Farrell et al., 2020; Kwok, Kacimi, et al., 2019). However, the Arctic summer melt season introduces complexities that contribute to uncertainties in its standard data products (Farrell et al., 2020; Magruder et al., 2025). As the snow cover melts, melt ponds form and evolve, creating spatially and temporally variable surface conditions (Arndt et al., 2025). These features dramatically alter the reflectance properties of the sea ice surface (Perovich & Polashenski, 2012), influencing the laser photon returns detected by ATLAS. The 532-nm wavelength can penetrate the water column, allowing the laser to detect returns from the bottom of melt ponds and capture their bathymetry in the ATL03 geolocated photon data product (Buckley et al., 2023; Farrell et al., 2020; Tilling et al., 2020). The ATL07 product, which provides sea ice surface height, includes a surface-type classification scheme to distinguish sea ice from leads and open water. Accurate classification is critical, as the ATL10 freeboard product depends on correctly identifying these surface types to determine ice freeboard. Because the same surface classification categories are used year-round, summer melt features—such as melt ponds or wet surfaces—may be misclassified or ambiguously labeled, reducing the accuracy of height retrievals and limiting the utility of ICESat-2 for summer sea ice thickness estimates.

Understanding how the melting sea ice surface affects ICESat-2 laser returns is essential to improve summer retrievals and enhance our ability to monitor the full seasonal cycle of Arctic sea ice thickness. In this study, we evaluate ICESat-2's performance in the presence of melt ponds using data from a targeted airborne validation campaign. By comparing high-resolution LiDAR and imagery collected under near-coincident overpasses, we assess the accuracy of ICESat-2 surface classifications and explore the potential for improving summer data products.

2. Study Area and Data

2.1. Summer Sea Ice Campaign and Flight Selection

The ICESat-2 summer sea ice airborne validation campaign was conducted in July 2022 to evaluate how surface melt affects the laser altimetry measurements over Arctic sea ice. The campaign aimed to (a) assess the performance of the ATL07 (sea ice height) and ATL10 (freeboard) products during summer melt, (b) improve their accuracy and precision under melt conditions, and (c) evaluate the feasibility of a melt pond depth retrieval algorithm for future incorporation into ICESat-2 standard products. Based out of Pituffik Space Base (formerly Thule Air Base), NASA's Gulfstream V aircraft conducted six science flights between July 11–26, 2022, each lasting 5–7 hr and targeting ICESat-2 orbits over a range of sea ice conditions including young ice, consolidated older ice, melt ponds at various drainage stages, and ridged ice (Saylam et al., 2025). High-resolution airborne RGBN imagery and LiDAR data were collected along ICESat-2 ground tracks with minimal temporal offset to evaluate the accuracy of satellite observations and derived products. Further information about the summer campaign and the preliminary assessment of data accuracy can be accessed in Saylam et al. (2025).

The first three flights of the campaign were separated from the ICESat-2 overpass by several hours, potentially limiting the quality of the freeboard analysis. Thus, we focus on a particularly high-quality flight characterized by clear skies, cloud-free aerial imagery, strong ICESat-2 returns, and minimal temporal offset between ICESat-2 and the airborne measurements (Figure 1a). On 26 July 2022, this flight was conducted over ICESat-2 Reference Ground Track 531, covering the strong ground tracks, GT2R and GT3R. The aircraft flew at ∼250 knots at an altitude of ∼500 m, collecting high-resolution LiDAR and imagery data from both the Chiroptera and Land Vegetation and Ice Sensor (LVIS) systems along a ∼300 km transect of sea ice north of Greenland and Ellesmere Island (Figure 2).

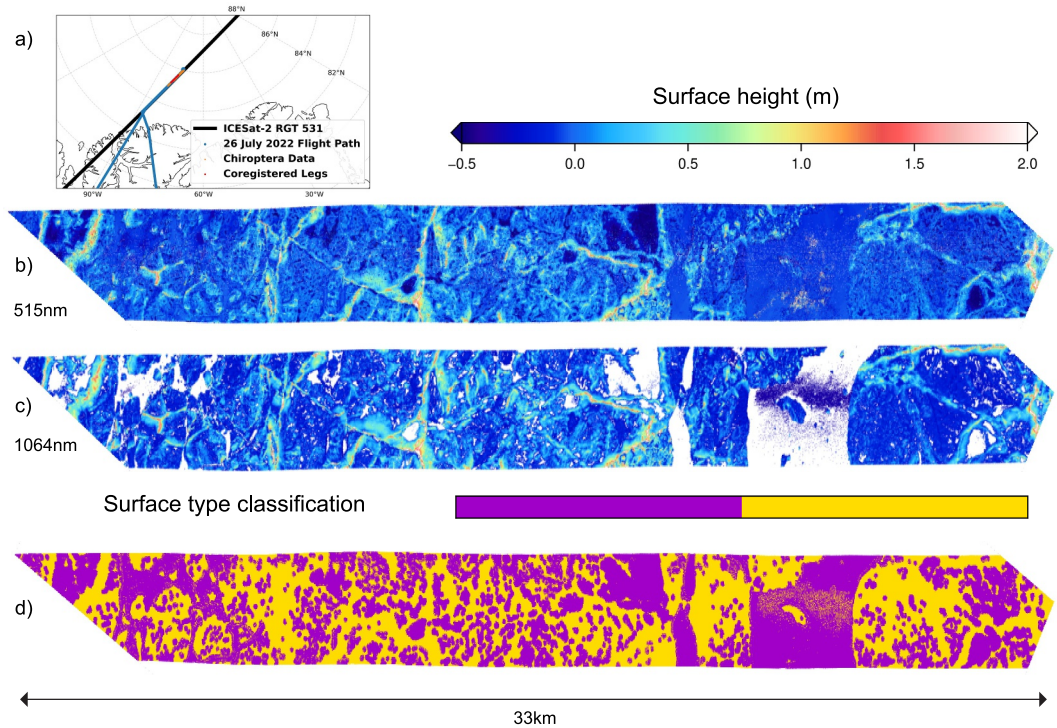


Figure 1. Chiroptera LiDAR Data. (a) Location of study showing ICESat-2 reference ground track (black), airborne flight path (blue), location of collected Chiroptera data (orange) and coregistered data (red). (b)–(c) Sea ice surface height shown for green laser (515 nm), and near infrared laser (1,064 nm). (d) Sea ice surface colored by Chiroptera surface type classification.

2.2. ICESat-2

The ICESat-2 photon-counting laser altimeter was launched on 15 September 2018. With a 92∘ inclination and coverage up to 88∘N, it is intended to measure changes in the polar region. ICESat-2 has 3 beam pairs separated by 3.3 km, with a "weak" and "strong" beam within each pair separated by 90 m. The ATL07 Sea Ice Height product provides sea surface and sea ice heights and surface types along each of the six beams. Each ATL07 height is derived from 150 photons in the ATL03 Global Geolocated Photon Height Product, resulting in varying ATL07 segment lengths ranging from ∼10–200 m in length for the strong beams.

Each segment is assigned a surface type based on statistics of the 150 aggregated photons (Kwok et al., 2025). Namely, the fitted Gaussian width of distribution of photons, the average photon rate, the sun elevation, and the background photon rate. ATL07 surface type classifications include cloud-covered segments (height_segment_type = 0), non-lead snow and ice surfaces (1), specular leads with low (2–3) or high (4–5) reflectance, dark smooth leads (6–7), dark rough leads (8–9), and unclassified or invalid segments (−1). During summer, categories 1–9 are further labeled with a "/pond" suffix, such as ice/pond or lead/pond. The sea surface height flag (ssh_flag) is used to identify segments that are suitable as sea surface reference height estimates for computing the local freeboard in ATL10. The sea surface height flag requires specular leads (types 2–5) whose height falls below the second percentile of sorted heights (Figure 2f).

2.3. Leica Chiroptera

The Leica Chiroptera 4X (CHIR) airborne LiDAR system includes green (515 nm) and near-infrared (1,064 nm) lasers (Saylam et al., 2025). The NIR channel captures surface elevations while the green wavelength can penetrate the water column enabling detection of both the sea ice surface and melt pond bathymetry. At an aircraft altitude of ∼500 m, the system provides a swath width of approximately 365 m with a nominal spatial resolution of ∼0.5 m. Geolocated surface elevations are derived by combining range measurements with GNSS/IMU-based position and orientation data. Surface type classifications are obtained using an automated algorithm that

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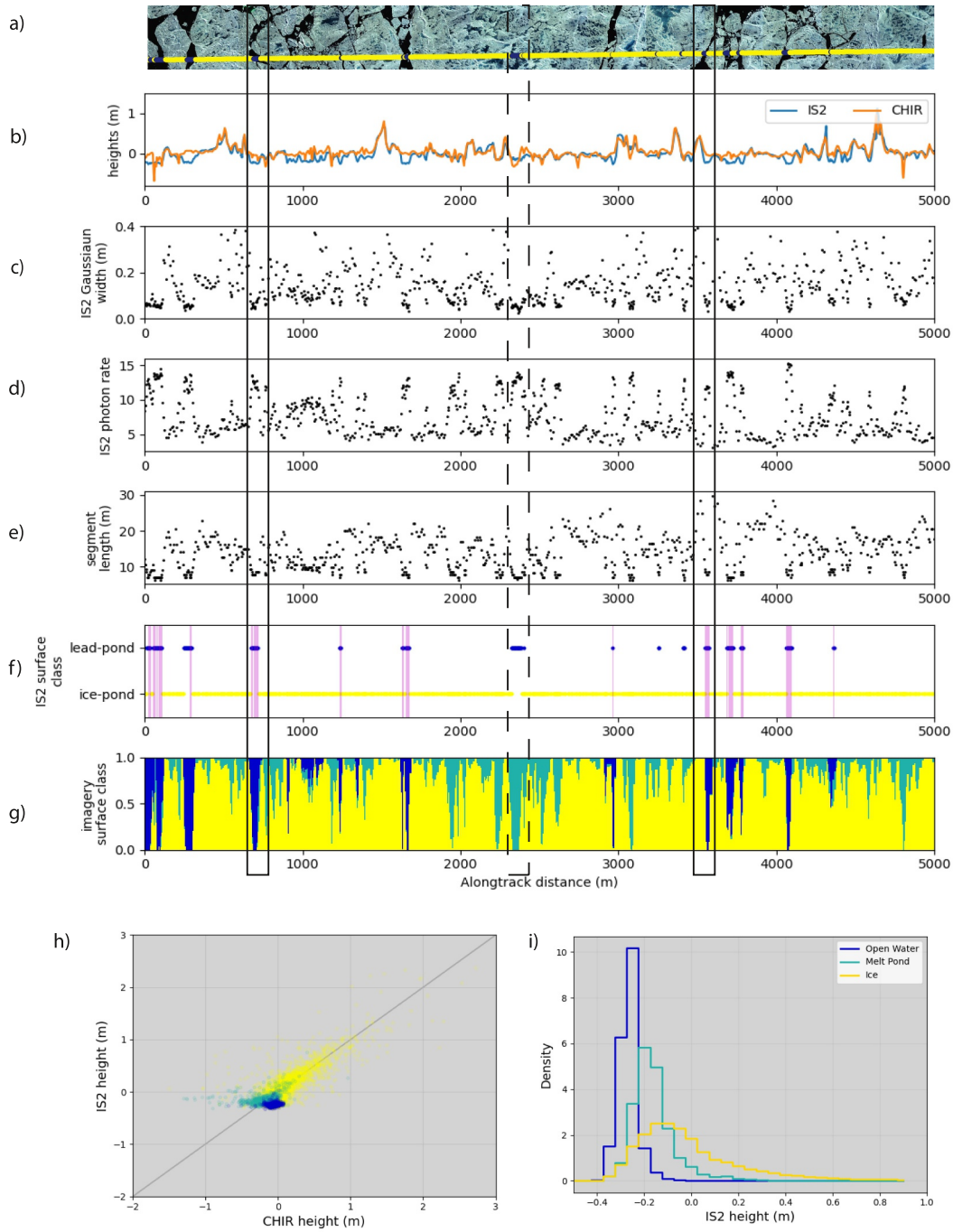


Figure 2. ICESat-2 and Chiroptera Data over a 5,000 m segment of melting Arctic sea ice captured 26 July 2022 (FL_1B). The sea ice surfaces throughout the data are classified as follows: ice (yellow), melt pond (teal), and open water (blue). The solid line box highlights an area of open water, the dashed boxes highlight areas containing melt ponds. (a) Chiroptera true color imagery with ICESat-2 ATL07 points overlain and colored by IS2 surface type classification. (b) surface heights from Chiroptera (green) (prior to the filtering described in Section 4.2.1) and ICESat-2 ATL07 (blue). (c) The width of the Gaussian fit of photons within each ICESat-2 ATL07 segment. (d) The ICESat-2 photon rate per shot averaged over the ATL07 segment. (e) ICESat-2 ATL07 segment length (f) ICESat-2 ATL07 surface type classification. The pink vertical lines indicates lead segments with the ssh_flag. (g) Fractional classification of Chiroptera image pixels within each ATL07 footprint. (h) ICESat-2 heights compared with Chiroptera heights colored by surface type verified by Chiroptera imagery. (i) ICESat-2 heights by surface type.

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incorporates return characteristics (e.g., intensity, echo width, return number) and spatial metrics (e.g., elevation variability and local geometry) to distinguish between sea ice and water surfaces and derive a classified surface height product (Figure 1d). The system also includes a multispectral camera acquiring imagery in red, green, blue, and near-infrared bands, with an actual pixel size of approximately 0.06 m, which were reduced to 0.5-m resolution for nominal analysis.

3. Pre-Processing

3.1. Airborne Imagery Classification and Characterization

Following Buckley et al. (2020, 2023), we apply an algorithm to the Chiroptera images to classify each individual pixel as sea ice, open water, or melt pond. For each image, we use the histograms of individual spectral channels (red, green, blue, and near-infrared) and specific band combinations. Natural breaks in these histograms allow us to distinguish surface types based on their spectral signatures. Analyzing the images individually ensures that the classification algorithm remains robust to variations in image brightness caused by changing overhead lighting conditions during the flight. We applied the image classification algorithm, based on methods from Buckley et al. (2020, 2023), to approximately 500 images collected along the flight line. We found a mean melt pond fraction of 15.1% (5.3%) and a mean sea ice concentration of 90.5% (9.3%). Sea ice in the survey domain was primarily composed of second-year and older sea ice types. The state of the surface conditions was characteristic of early-to-mid summer, with snow drifts present along ridges and melt ponds in an advanced stage of evolution (Eicken et al., 2002). There was roughly an equivalent number of ponds that were interconnected versus isolated, with the pond geometry depending on the surface relief. Some ponds were not well developed, indicating earlier stages of pond evolution; these were observed on thinner, smoother sea ice. On multiyear ice, old (with snow drifts) and new (with no snow drifts) ridges were present, and ponds tended to be more interconnected and developed. In these areas, many ponds appeared to have drained or were in the process of drainage. There were no obvious signs of freezing, as both small and large ponds were open with no ice lids. The largest ponds occurred in "basins" surrounded by heavily deformed, thick sea ice and had not undergone drainage. Some areas also had sediment-laden ice.

3.2. Coregistration of Chiroptera and ICESat-2 Height Profiles

Due to local ice drift conditions and uncertainties in the predicted locations of the two altimeters, the CHIR and ICESat-2 height samples are not expected to be exactly coregistered on the surface. Here, we follow the approach described in Kwok, Kacimi, et al. (2019) and refine the coregistration by maximizing the correlation between the ICESat-2 height samples (in ATL07) and candidate Chiroptera height profiles at the ICESat-2 sample locations. Candidate height profiles from the Chiroptera data are constructed by shifting ICESat-2 sample locations (in 10-km segments) by small increments in the along-track and cross-track directions and slight tilts from the middle of the 10-km segment. At each shift/tilt increment, weighted averages of the Chiroptera heights within each ICESat-2 segment are computed. Because the number of Chiroptera samples associated with each ICESat-2 location varies—since the spatial resolution of the surface heights in ATL07 varies along track—the ICESat-2 footprint is modeled as a Gaussian with a full width of 14 m to correctly weight the Chiroptera samples within a given ATL07 along-track segment. In total, five 10-km segments were successfully co-registered, three within FL1-B and two within FL1-C. The results are discussed in Section 4.2.

4. Comparison of ICESat-2 and Chiroptera

4.1. Surface Type Classification

Along a ∼50 km stretch of data collected on 26 July 2022, we analyzed 7,593 ATL07 segments: 6,784 were classified as ice/pond, 800 as specular lead/pond, and 9 as dark lead/pond. To compare and validate ATL07 surface type classifications against those derived from imagery, we sampled each ATL07 segment footprint, stadium-shaped regions defined by the 11 m width of the ICESat-2 beam and the ATL07 segment length parameter, with rounded ends. Due to the varying segment length of ATL07, the number of pixels within a footprint is varied, but on average, the footprint covers an area of 161 m^2 and contains 646 Chiroptera pixels with varying surface type classifications. The "ground truth" classification is assigned based on the surface type that occupies the largest proportion of pixels within a given footprint.

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Table 1
Confusion Matrix Comparing ICESat-2 Surface Type Classifications (Rows) With Image-Based Classifications (Columns)

		Image-based classification			
		Open water	Melt pond	Ice	Total
ICESat-2	Open Water (ssh_flag)	554 (288)	110 (4)	144 (86)	808 (379)
	Melt Pond	0	0	0	0
	Ice	263	849	5,550	6,662
	Total	817	959	5,694	7,470

Note. Totals are slightly less than the total ICESat-2 segments mentioned in the text as some segments do not line up with classified imagery.

For comparison with image-based classifications, we group all lead/ponds (categories 2–9) as open water, and treat ice/pond segments (category 1) as ice (see Section 2.2). The resulting confusion matrix (Table 1) compares ICESat-2 surface type classifications to those derived from high-resolution imagery. ICESat-2 identifies ice with 97% accuracy and open water with 68% accuracy. We note that some misclassifications are likely due to inaccuracies in the drift correction, not necessarily the ICESat-2 surface classification algorithm. Melt ponds in ICESat-2 data are divided between ice and open water ATL07 classifications: approximately 11.5% of melt pond segments are labeled as open water in ATL07, while 88.5% are classified as ice in ATL07. Of the 379 segments flagged for use as sea surface reference (ssh_flag = 1), 288 were classified as open water in the coincident imagery, while 86 were classified as ice and 4 as melt pond (Table 1). Of the 959 melt pond returns as classified by the imagery, 110 were classified as lead/pond (11%) and only 4 were flagged as sea surface height candidates for the ATL10 calculation. Despite moderate accuracy in distinguishing open water, the implications for ATL10 freeboard retrieval depend on whether misclassified surfaces are used as sea surface height references and this is discussed further in Section 5.

4.2. Surface Heights and Freeboards

Because increases in air temperature and surface melt can lead to deeper penetration of the ICESat-2 532 nm signal into the snow layer or melt ponds, we provide here an assessment of ICESat-2 surface heights and freeboards through a comparison with Chiroptera acquisitions.

4.2.1. Comparison of Surface Heights

While the green channel of the Chiroptera LiDAR (515 nm) is similar in wavelength to that of ICESat-2 (532 nm), the two systems rely on fundamentally different technologies to retrieve surface heights. Chiroptera records the full return waveform of a beam and identifies surface and bathymetric elevations based on peak detection. As a bathymetric LiDAR system, its green channel is designed to capture returns originating throughout the water column, including interactions with subsurface features. In conditions where the surface return is weak or where volume scattering is significant, the detected peak may therefore correspond to a stronger subsurface contribution rather than the air–snow (or air-water for melt ponds) interface. A stronger subsurface return relative to the surface return leads to greater error. In contrast, ICESat-2 operates a photon-counting LiDAR, and its sea ice surface-finding algorithm mitigates the effects of subsurface scattering by windowing the photon height distributions and removing delayed photons from the final distributions. This enables a more accurate representation of the returns that originate near the surface. Although the Chiroptera NIR channel offers limited penetration compared to the 515 nm due to absorption and scattering it could, in principle, be used to assess ICESat-2 surface heights and freeboard. In practice, the low signal-to-noise ratios result in incomplete coverage, particularly over leads and at high grazing angles. Since freeboard estimation requires the determination of local sea level from leads, the NIR-derived water surface elevations were therefore unreliable for this purpose. In the following analysis, we use only the heights from the 515 nm channel, which we refer to as Chiroptera for simplicity.

To make Chiroptera data comparable with ICESat-2, we first select Chiroptera-derived heights associated with two surface types: water surface (constrained by retrievals from both the 515 nm and NIR channels) and ground, corresponding to the first and only return from hard surfaces. During the coregistration procedure, we find that removing the lowest 40th percentile of Chiroptera height samples within each ICESat-2 segment improves the

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correlation between the two data sets. This filtering preferentially retains surface elevation retrievals while excluding anomalously low elevations that may result from subsurface scattering, bathymetric returns, or erroneous surface detection (Figures 1d and 2b). The comparison of surface heights between ICESat-2 and Chiroptera for five coregistered 10-km segments is summarized in Figure 3. For ICESat-2, all segments are associated with Beam 1 (GT3R, strong beam). The regression analysis (slope, correlation, and standard error) between the surface heights supports the assessment of the coregistration procedure and the ICESat-2 summer heights. Overall, all coregistered segments are strongly correlated ($R \geq 0.9$). Regression slopes range from 0.84 to 0.96, indicating good agreement between the two data sets, with a slight underestimation of ridge heights by ICESat-2 relative to Chiroptera. Intercepts are small (0.002–0.04 m), suggesting minimal systematic offset.

4.2.2. Comparison of Freeboards

An additional motivation for the 2022 summer campaign was the opportunity to assess, for the first time, summer freeboard estimates from ICESat-2. In this analysis, we calculate freeboard using two different approaches. The first (h^1_{fS2} and h^1_{fCh}) uses the lowest surface height —from each sensor separately— within each 10-km segment as an estimate of local sea level. The second uses the average height of samples identified as specular leads in the ICESat-2 ATL07 product (h^2_{fS2} and h^2_{fCh}). As shown in Figure 3, returns from specular leads identified by ICESat-2 are available for all five co-registered segments.

The first freeboard estimate provides an assessment of the coregistration quality. Over the five 10-km segments, the freeboard differences ($h^1_{fS2} - h^1_{fCh}$) range from -0.05 ± 0.12 m to 0.01 ± 0.10 m. Differences are lower (−0.01 m on average) for segments 1–3 which are associated with the FL1-B Chiroptera data set. Except for segment 4, where the coregistration quality is lower, as indicated by the regression analysis, the minimum heights from ICESat-2 and Chiroptera along the 10-km profiles agree within 1 cm. This is an indication that h^1_{fS2} and h^1_{fCh} are relatively well aligned, especially for the FL1-B Chiroptera flight portion. FL1-B occurred closer in time to the ICESat-2 pass, with a lag of 12–17 min, whereas FL1-C occurred 17–21 min after the pass, allowing for greater sea ice drift.

To derive h^2_{fS2} and h^2_{fCh}, we first compare height differences at the locations of ICESat-2 specular leads. We find a mean bias of -0.09 ± 0.01 m, with Chiroptera heights consistently higher than those from ICESat-2. This is evident in the line plots and scatterplots displayed in Figure 3. The better agreement over ice floes (as indicated by the relatively low differences between h^1_{fS2} and h^1_{fCh}) suggests that the height differences over leads are likely caused by a surface-type-dependent effect. Particularly over specular leads, the use of peak detection in the Chiroptera processing may bias elevations toward an earlier part of the return, yielding higher elevations than ICESat-2. Over ice floes, where the return is more diffuse, this bias is reduced and the two systems agree more closely. Therefore, a bias correction over leads is applied to align the Chiroptera reference surface heights with those of ICESat-2. Resulting differences between h^2_{fS2} and h^2_{fCh} vary between $-0.01 \pm 0.11m$ and $0.01 \pm 0.09m$ for segments 1–3 and $0.02 \pm 0.11m$ and $0.03 \pm 0.09m$ for segments 4–5. Assuming that the ICESat-2 specular leads are correctly identified (see above discussion on surface classification), the ICESat-2 and Chiroptera freeboards are relatively unbiased.

5. Discussion

In Figure 2, we show a 5 km section of data featuring leads and melt ponds. Within this transect, we identified sections corresponding to open water (solid boxes) and melt ponds (dashed box) (Figure 2). Open water segments were typically classified as specular leads by ATL07 (Figure 2, solid box) and were flagged for sea surface height references in the calculation of the sea ice freeboard product, ATL10. These segments exhibited classic lead signatures: low elevation, narrow vertical spread in photon returns, and high photon rates. In contrast, melt pond segments (Figure 2, dashed box), although also classified as specular leads, were not used as sea surface height references for calculation of ATL10. While their return signatures closely resembled those of leads, they appeared at slightly higher elevations, indicating that despite classification similarities, the ATL07 algorithm requires segments with heights below the second percentile of all sorted heights for a positive sea surface height flag, thus excluding them from sea level reference calculations. The results are consistent with Tilling et al. (2020) who compared ICESat-2 returns over melt ponds with near-coincident high resolution satellite imagery from Sentinel-

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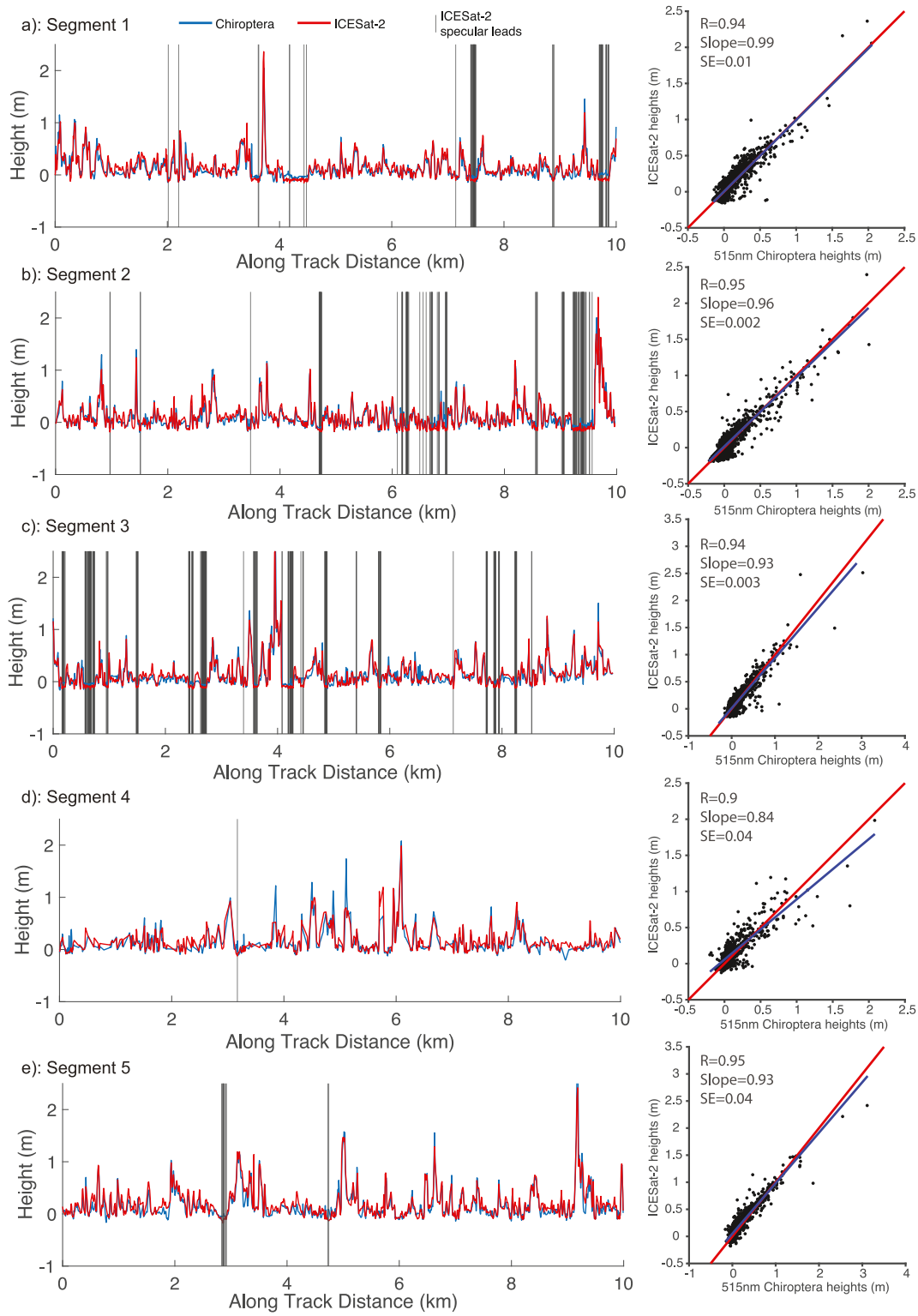


Figure 3. Comparison of ICESat-2 and Chiroptera (515 nm) surface heights in 10-km segments. (a–e) Along-track surface profiles (10-km segments) and corresponding scatterplots with regression results (slope, correlation, and standard error) for five segments along the flight tracks on 26 July 2022.

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2 (10 m) and WorldView-2 (1.84 m) in the Central Arctic Ocean and Canadian Archipelago. They found melt ponds typically produce quasi-specular returns (>16 photons per return) and exhibit higher elevations than leads. They also identified a large melt pond with lower reflectivity, which they attributed to increased wind-induced surface roughness. In contrast, the melt ponds identified in our study all produced specular returns. This difference may reflect the smaller, more sheltered melt ponds sampled here, which are less susceptible to wind roughening because of their shorter fetch.

The analyzed data show that some melt pond surfaces are classified as lead/pond, as expected because the ICESat-2 classification scheme does not include a separate melt pond category. Because the ssh flag requires segments to fall within the lowest 2% of heights, the melt ponds flagged as ssh candidates are very close to the local sea level and are unlikely to have a significant negative impact on the freeboard calculation.

The ice-category false positives, though more numerous (86 of 379), likewise do not meaningfully degrade the sea surface height estimate, for two reasons. First, the falsely flagged segments are predominantly mixed-surface segments rather than pure ice: their footprints contain a median of 22% open-water pixels, and only 9 of 86 are more than 95% ice-covered, suggesting lead edges and refrozen leads rather than floe interiors. Some of these apparent false positives may also not be misclassifications at all, but rather the consequence of drift-correction errors or small-scale floe and brash ice motion between the ICESat-2 and airborne acquisitions. Second, their retrieved heights are nearly identical to those of the correctly identified open-water segments (-0.258 ± 0.024 m vs. -0.253 ± 0.023 m), which the coincident Chiroptera LiDAR confirms (median surface heights of 0.23 vs. 0.21 m): the ice-classified segments carrying the ssh flag lie at the local water level rather than at floe height. Consequently, including all falsely flagged segments changes the derived sea surface height by less than 0.2 cm, well within the ATL07 height uncertainty.

These results suggest that ATL10 remains reliable during summer conditions. The data set is limited, however, representing only 50 km of observations over multiyear ice in July. Variations in ice type and atmospheric conditions may influence how well ICESat-2 can distinguish leads from melt ponds, and additional observations will be needed to assess performance across a broader range of conditions.

Figure 2h shows the correlation between ICESat-2 heights and the unfiltered Chiroptera height retrievals from all coincident ICESat-2 and Chiroptera segments that met our minimum correlation criteria. The unfiltered data shown here illustrate why filtering is necessary, especially for melt pond returns. Melt pond points (teal) show a broad spread and consistent deviation from the ICESat-2 heights. This variability occurs because the unfiltered melt pond heights include contributions from several surfaces, such as pond bathymetry, the water surface, and additional reflections and scattering within the pond. The distribution of these same unfiltered heights appears in Figure 2i. ICESat-2 lead heights form a narrow and distinct mode, while melt pond and ice surfaces span wider elevation ranges. The melt pond distribution is centered at lower elevations than the surrounding ice and slightly higher than the lead mode, although the distributions show some overlap.

6. Conclusions

Although ICESat-2 summer sea ice products do not include a dedicated classification for melt ponds, we find that over the $\sim$ 50 km multiyear ice transect analyzed here, melt ponds have a negligible impact on freeboard estimates. The ssh flag effectively filters out higher-elevation segments classified as leads that are likely melt ponds. As a result, the reference sea surface elevations used in the freeboard calculation correspond to leads or melt ponds that are hydrologically connected to the underlying ocean and therefore are at the height of local sea surface elevation. In fact A. Petty et al. (2025) used ICESat-2 summer freeboard and a snow model to estimate summer sea ice thickness, showing good agreement with both CryoSat-2 summer sea ice thickness estimates and the airborne IceBird electromagnetic (EM-bird) thickness retrievals.

In addition to refining surface classification, future work will examine potential centimeter-scale biases in the ICESat-2 freeboard retrieval algorithm using coincident airborne observations, including the 2 August 2024 NASA ARCSIX flight, which collected simultaneous camera imagery and LVIS data (Taylor, 2024). A key focus is the influence of highly variable summer surface photon rates (Figure 2). Preliminary comparisons with the release 7 ATL07 fixed 10-m heights (Wimert et al., 2025) indicate reduced residual biases in both height and freeboard retrievals. We will further investigate whether regions with elevated photon return rates are preferentially biased by the 150-signal-photon retrieval algorithm currently used in ATL10, consistent with a possible electromagnetic bias.

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Building on the efforts from this study and ongoing work in the community (e.g., A. A. Petty et al., 2021), there is potential to compile a database of high resolution imagery and ICESat-2 observations. The work of Liu et al. (2025) demonstrated that combining coincident ICESat-2 observations with Sentinel-2 multispectral imagery enables improved discrimination of leads through a machine-learning classification approach. With the application of surface type classification and drift correction to the imagery, this study could also serve as a training data set to refine the ICESat-2 surface type classifications, allowing the inclusion of melt ponds and potentially other types of sea ice surface types with distinct radiometric signatures.

AI Tools Disclosure

EB acknowledges use of Generative AI (ChatGPT-5.3) for coding in figure creation and for assistance with sentence restructuring.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The airborne LiDAR data set used in this study is available to the public at Texas Data Repository (Saylam, 2024): https://doi.org/10.18738/T8/MYOUUV and the NIR data are here (Saylam et al., 2026): https://doi.org/10.5067/B0EV8BIL17U2. The ICESat-2 ATL07 release 6 data may be found here: https://nsidc.org/data/atl07/versions/6. The image classification code can be found here: https://github.com/ellenbuckley/IS2_CHIR. The coregistered ICESat-2 data and CHIR green LiDAR for the 26 July 2022 flight are archived here: https://doi.org/10.5281/zenodo.21364639.

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